

## Chhattisgarh Swami Vivekananda Technical University, Bhilai (C.G.)

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|--------------------------------------------------------------------|---------------------------------------------------|
| Program / Semester: <b>B.Tech (IV)</b>                             | Branch: <b>Computer Science &amp; Engineering</b> |
| Subject: <b>Database Management System</b>                         | Course Code: B022413(022)                         |
| Total / Minimum-Pass Marks (End Semester Exam):<br><b>100 / 35</b> | L: 3 T: 1 P: 0 Credits: 4                         |
| Class Tests & Assignments to be conducted: 2 each                  | Duration (End Semester Exam): <b>03 Hours</b>     |

**UNIT I Database system architecture:** Data Abstraction, Data Independence, Data Definition Language (DDL), Data Manipulation Language (DML). **Data models:** Entity-relationship model, network model, relational and object-oriented data models, integrity constraints, data manipulation operations.

**UNIT II Relational query languages:** Relational algebra, Tuple and domain relational calculus, SQL3, DDL and DML constructs, Open source and Commercial DBMS - MYSQL, ORACLE, DB2, SQL server. **Relational database design:** Domain and data dependency, Armstrong's axioms, Normal forms, Dependency preservation, Lossless design.

**UNIT III Query processing and optimization:** Evaluation of relational algebra expressions, Query equivalence, Join strategies, Query optimization algorithms. **Storage strategies:** Indices, B-trees, hashing,

**UNIT IV Transaction processing:** Concurrency control, ACID property, Serializability of scheduling, Locking and timestamp-based schedulers, Multi-version and optimistic Concurrency Control schemes, Database Recovery.

**UNIT V Database Security:** Authentication, Authorization and access control, DAC, MAC and RBAC models, Intrusion detection, SQL injection. **Advanced topics:** Object oriented and object relational databases, Logical databases, Web databases, Distributed databases, Data warehousing and data mining.

### Text books:

1. "Database System Concepts", 6th Edition by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill.
2. "Fundamentals of Database Systems", 5th Edition by R. Elmasri and S. Navathe, Pearson Education
3. "Introduction to Database Systems", 8th Edition by C J Date, Addison Wesley, 2003.

### Suggested reference books

1. "Principles of Database and Knowledge – Base Systems", Vol 1 by J. D. Ullman, Computer Science Press.
2. "Foundations of Databases", Reprint by Serge Abiteboul, Richard Hull, Victor Vianu, Addison-Wesley

### Course Outcomes [The students should be able to]:

1. Be familiar with basic concepts of RDBMS, Relational data model & be able to write relational algebra expressions for queries;
2. Be familiar with basic database storage structures and access techniques: file and page organizations, indexing methods including B-tree and hashing;
3. Understand DML, DDL and will be able to construct queries using SQL by knowing the importance of data & its requirements in any applications;
4. Utilize a database modelling technique for a single entity class, a one-to-one (1:1) relationship between entity classes, a one-to-many (1:M) relationship between entity classes, a many-to-many (M:M) relationship between entity classes, and recursive relationships;
5. Be familiar with the basic issues of transaction, its processing and concurrency control.



**B022413(022)**

**B.Tech. (Fourth Semester) Examination  
April-May 2023**

**(New Scheme)**

**(CSE Branch)**

**DATABASE MANAGEMENT SYSTEM**

***Time Allowed : Three hours***

***Maximum Marks : 100***

***Minimum Pass Marks : 35***

***Note : Attempt all questions. Part (a) is compulsory from each unit. Attempt any **two** parts from (b), (c) and (d) in each unit. All units carries equal marks.***

**Unit-I**

1. (a) Discuss data indepdence and its types in brief. 4

- (b) Explain Entity-relationship model with suitable example.
- (c) What is data abstraction? Describe various levels of data abstraction.
- (d) Define data model. Discuss different types of data models with proper example.

### Unit-II

2. (a) Define relational model and RDBMS. 4
- (b) Briefly explain about domain relational calculus and tuple calculus 8
- (c) Explain the following operations in relational algebra Selection, Projection, Renaming & Join. 8
- (d) Define normalization. Explain 1NF and 2NF with suitable example. 8

### Unit-III

3. (a) Discuss Join strategies in brief. 4

- (b) Define Query optimization and briefly explain about its algorithms with example. 8
- (c) Explain Hashing with example in detail. 8
- (d) Describe B+ trees in detail. 8

### Unit-IV

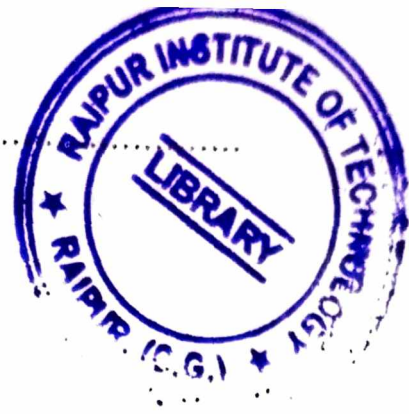
4. (a) Define Transaction & Database recovery. 4
- (b) What you mean by transaction processing? Explain states of transaction processing. 8
- (c) Explain ACID property. 8
- (d) Explain the shadow paging and online backup technique used for recovery of database. 8

### Unit-V

5. (a) Differentiate between data warehousing and data mining. 4
- (b) Describe Access Control in detail. 8

(c) Write short notes on Intrusion detection and SQL injection. 8

(d) Explain Distributed database with example. 8



**B019412(019)**

**B. Tech. (Fourth Semester) Examination,  
April-May 2023**

**(New Scheme)**

**(Chemical Engg. Branch)**

**HEAT TRANSFER OPERATION**

***Time Allowed : Three hours***

***Maximum Marks : 100***

***Minimum Pass Marks : 33***

***Note : Attempt all questions. Part (a) of each question is compulsory and carries 4 marks. Solve any **one** part from part (b) & (c) and carries 16 marks each.***

**Unit-I**

1. (a) Define Conductivity. Give examples of materials having high and low conductivity.

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[ 2 ]

- (b) A furnace wall made up of steel plate 10 mm thick lined on inside with silica brick 150 mm thick on the outside with magnesite brick 150 mm thick. The temperature on inside edge of the wall is 973 K and on the outside is 288 K. Calculate the quantity of heat passed in watts per  $m^2$ . It is required to reduce the heat flow to  $1163 \text{ W/m}^2$  by means of air gap between steel plate and magnesite brick. Estimate the width of this gap. Thermal conductivities in  $\text{W/(m K)}$  are 16.86, 1.75, 5.23 and 0.033 respectively for steel, silica brick, magnesite brick and air. 16
- (c) A steam pipeline, 150/160 mm in diameter, carries steam. The pipeline is lagged with a layer of heat insulating material ( $k = 0.08 \text{ W/(mK)}$ ) of thickness 100 mm. The temperature drops from 392.8 K ( $199.8^\circ\text{C}$ ) to 313 K ( $40^\circ\text{C}$ ) across the insulating surface. Determine the rate of heat loss per 1 m length of pipe line. 16

### Unit-II

2. (a) Define natural convection. Give one-one example of natural and forced convection. 4

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- (b) In a double pipe counter current flow heat exchanger, 10000 kg/h of an oil having a specific heat of  $2095 \text{ J/(kgK)}$  is cooled from 353 K ( $80^\circ\text{C}$ ) to 323 K ( $50^\circ\text{C}$ ) by 8000 kg/h of water entering at 298 K ( $25^\circ\text{C}$ ). Calculate the heat exchanger area for an overall heat transfer coefficient of  $300 \text{ W/(m}^2\text{K)}$ . Take  $C_p$  for water as  $4180 \text{ J/(kg K)}$ . 16
- (d) Calculate the heat transfer area of 1-2 heat exchangers from the following data : Inlet and outlet temperature of hot fluid are 423 K ( $150^\circ\text{C}$ ) and 353 K ( $80^\circ\text{C}$ ) respectively. Inlet and outlet temperatures of cold fluid are 303 K ( $30^\circ\text{C}$ ) and 318 K ( $45^\circ\text{C}$ ) respectively.
- Overall heat transfer coefficient =  $4100 \text{ W/(m}^2\text{K)}$   
Heat Loss = 407 kW  
L.M.T.D. Correction factor = 0.84 16

### Unit-III

3. (a) What is pool boiling? Give an example for it. 4
- (b) Explain in detail : 16

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- (i) Film wise condensation  
 (ii) Dropwise condensation  
 (c) Calculate the rate of heat loss from a 6 m long horizontal steam pipe, 60 mm o.d. when carrying steam at 800 kN/m<sup>2</sup>. The temperature of atmosphere and surrounding is 290 K.

Data : Take emissivity,  $e = 0.85$  and Stefan-Boltzman Constant,

$$\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2 \cdot \text{K}^4)$$

Fil coefficient ( $h_c$ ) for heat loss by natural convection can be calculated by :

$$h_c = 1.64 (\Delta T)^{0.25} \text{ W/(m}^2 \cdot \text{K)}$$

Steam is saturated at 800 N/m<sup>2</sup> and 443 K (170°C) 16

#### Unit-IV

4. (a) Differentiate between single pass and multi pass shell tube heat exchanger. 4

- (b) Hot oil at a rate of 1.21 g/s [ $C_p = 2083 \text{ J/(kg} \cdot \text{K)}$ ] flow through double pipe heat exchanger. It enters at 633 K (360°C) and leaves at 573 K (300°C).

The cold fluid enters at 303 K (30°C) and leaves at 400 K (127°C). If the overall heat transfer coefficient is 500 W/(m<sup>2</sup>·K), calculate the heat transfer area for (i) parallel flow and (ii) countercurrent flow. 16

- (c) With the help of neat diagram explain any one heat exchanger equipment. 16

#### Unit-V

5. (a) Define steam economy. 4

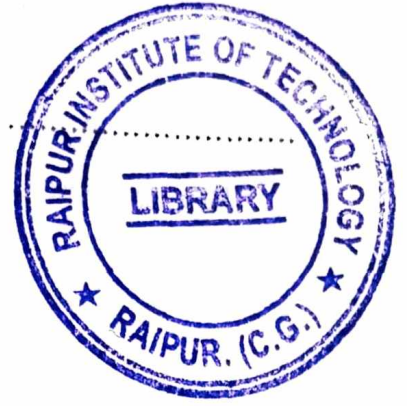
- (b) An evaporator is operating at atmospheric pressure. It is desired to concentrate the feed from 5% solute to 20% solute at a rate of 5000 kg/h. Dry saturated steam at a pressure corresponding to saturation temperature of 399 K is used. The feed is at 298 K and boiling rise is 5 K. Overall heat transfer to be provided. 16

**Data :** Treating solution as pure water, latent heat of condensation of steam at 399 K = 2185 kJ/kg  
 Latent heat of vaporization of water = 2257 kJ/kg  
 Specific heat of feed = 4.187 kJ/(kgK)

- (c) Explain multiple effect evaporators with neat & clean diagram.

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**B022413(022)**

**B.Tech. (Fourth Semester) Examination**

**Nov.-Dec. 2023**

**(CSE Branch)**

**DATABASE MANAGEMENT SYSTEM**

*Time Allowed : Three hours*

*Maximum Marks : 100*

*Minimum Pass Marks : 35*

*Note : Attempt all questions. Attempt any **two** parts from (a), (b) and (c) of each unit. Each part carries 10 marks.*

**Unit-I**

1. (a) Explain ANSI/SPARC 3 level architecture of DBMS.

- (b) Explain different type of data models.
- (c) Explain different types of Key in relational model with examples.

### Unit-II

2. (a) What are the different types of functional dependencies? Consider the following set of FD's :

$AB \rightarrow C$

$A \rightarrow DE$

$B \rightarrow F$

$F \rightarrow GH$

$D \rightarrow IJ$

Find :

- (i) super key
  - (ii) irreducible set
  - (iii) candidate key
- (b) Explain Normalization with examples.

- (c) Explain the following functions :
- (i) Select operation
  - (ii) Project Operations
  - (iii) Union
  - (iv) Intersection
  - (v) Rename

### Unit-III

3. (a) Explain Query optimization. Explain the structure of query optimizer.
- (b) Explain different approaches of storage strategies in detail.
- (c) How to evaluate relational algebra expressions in DBMS.

### Unit-IV

4. (a) Give the significance of serializability in database transaction. How view serializability is different from conflict serializability.

- (b) What is lock compatibility matrix? Explain with an example the concurrency control techniques for scheduling database transactions.
- (c) Discuss the time stamp ordering protocol for concurrency control. How does strict time stamp ordering differ from basic time stamp ordering.

### **Unit-V**

5. (a) Explain the architecture of knowledge discovery using data warehousing and mining.
- (b) What is logical and web database? Differentiate between object oriented and object relational database.
- (c) Write short notes on :
- (i) Intrusion detection techniques in database
  - (ii) Authorization, Authentication and access control in database



**B022413(022)**

**B.Tech. (Fourth Semester) Examination  
April-May 2024**

**(CSE Branch)**

**DATABASE MANAGEMENT SYSTEM**

***Time Allowed : Three hours***

***Maximum Marks : 100***

***Minimum Pass Marks : 35***

***Note : Attempt all questions. Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d).***

**Unit-I**

**1. (a) Define RDBMS.**

**4**

[ 2 ]

- (b) What are the various type of data model? Explain brief. 8
- (c) Explain DBMS system structure and its components. 8
- (d) Consider library maintains data about books. It is define as. 8
- (i) Entity book with attributes year title, price and ISBN.
- (ii) Entity author with attributes name, address, email.
- (iii) Construct E-R diagram showing relation between book and author and between book publisher

#### Unit-II

2. (a) Define SQL. 4
- (b) Define Relational Algebra. What are the operation in relational algebra? 8
- (c) Explain tuple relation calculus and domain relation calculus? 8
- (d) Explain Armstrong's rule for functional dependencies with example. 8

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#### Unit-III

3. (a) Define Query processing. 4
- (b) What is Query equivalence in DBMS? 8
- (c) Explain indices and B-tree. 8
- (d) Join strategies with example. 8

#### Unit-IV

4. (a) Define database recovery. 4
- (b) How the view and conflict serializability support the execution of concurrent transaction? Explain with suitable example. 8
- (c) Describe ACID properties of transaction. 8
- (d) List all the locking techniques for concurrency control over real time transaction. 8

#### Unit-V

5. (a) Define DAC. 4
- (b) Differences between object oriented and object relational database. 8

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(c) What is role based access control? 8

(d) Write short notes on : (any two) 8

(i) MAC

(ii) Datawarehousing and data mining

(iii) Web mining



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**B.Tech. (Fourth Semester) Examination**

**April-May 2021**

**(CSE Branch)**

**DATABASE MANAGEMENT SYSTEM**

***Time Allowed : Three hours***

***Maximum Marks : 100***

***Minimum Pass Marks : 35***

***Note : Attempt all questions. Attempt any two parts from (a), (b) and (c) of each unit. Each part carries 10 marks.***

**Unit-I**

1. (a) Explain the purpose of database systems. Differentiate DBMS versus file system.

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[ 2 ]

- (b) Discuss three levels architecture in DBMS.
- (c) Write about Relational Model. Also explain the concept of keys with the help of suitable example.

### Unit-II

2. (a) Explain join operators used in Relational Algebra.
- (b) Consider the relational database

employee(person-name,street,city)

works(person-name,company-name,salary)

company(company-name,city)

managers(person-name,manager-name).

Give an expression in the relational algebra for each request :

- (i) Modify the database so that Jones now lives in Newtown.
- (ii) Give all employees of First Bank Corporation a 10 percent salary raise.
- (iii) Give all managers in this database a 10 percent salary raise.

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- (iv) Give all managers in this database a 10 percent salary raise, unless the salary would be greater than \$100,000. In such cases, give only a 3 percent raise.
  - (v) Delete all tuples in the works relation for employees of Small Bank Corporation.
- (c) Define Normalization. Explain 3NF and BCNF.

### Unit-III

3. (a) Explain the steps in query processing by giving appropriate example.
- (b) What is evaluation plan and equivalence rules in Query optimization.
- (c) Write notes on : (any **two**)
- (i) Indices
  - (ii) B-Trees
  - (iii) Hashing

### Unit-IV

4. (a) Define Transaction. Write ACID properties of Transaction.

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[ 4 ]

- (b) What are Concurrent Schedules? Write in detail about serializability.
- (c) What is concurrency control mechanism? Explain Multi-version and Concurrency Control scheme.

### Unit-V

- 5. (a) What are the three pillars of security? Explain DAC, MAC and RBAC.
- (b) What is SQL Injection? How SQL Injection works and its defense mechanism?
- (c) Write notes on : (any **two**)
  - (i) Web databases
  - (ii) Distributed databases
  - (iii) Data warehousing and data mining

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**B022413(022)**

**B.Tech. (Fourth Semester) Examination**

**Nov.-Dec. 2021**

**(CSE Branch) A/C T E**

**DATABASE MANAGEMENT SYSTEM**

***Time Allowed : Three hours***

***Maximum Marks : 100***

***Minimum Pass Marks : 35***

***Note : Attempt all questions. Attempt any two parts from (a), (b) and (c) of each unit. Each part carries 10 marks.***

**Unit-I**

1. (a) Explain ANSI/SPARC 3 level architecture of DBMS.

[ 2 ]

- (b) Explain different type of data models.
- (c) Explain different types of Key in relational model with examples.

### Unit-II

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$AB \rightarrow C$

$A \rightarrow DE$

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$F \rightarrow GH$

$D \rightarrow IJ$

Find :

- (i) super key
  - (ii) irreducible set
  - (iii) candidate key
- (b) Explain Normalization with examples.

[ 3 ]

- (c) Explain the following functions :
- (i) Select operation
  - (ii) Project Operations
  - (iii) Union
  - (iv) Intersection
  - (v) Rename

### Unit-III

3. (a) Explain Query optimization. Explain the structure of query optimizer.
- (b) Explain different approaches of storage strategies in detail.
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### Unit-IV

4. (a) Give the significance of serializability in database transaction. How view serializability is different from conflict serializability.

[ 4 ]

- (b) What is lock compatibility matrix? Explain with an example the concurrency control techniques for scheduling database transactions.
- (c) Discuss the time stamp ordering protocol for concurrency control. How does strict time stamp ordering differ from basic time stamp ordering.

#### **Unit-V**

- 5. (a) Explain the architecture of knowledge discovery using data warehousing and mining.
- (b) What is logical and web database? Differentiate between object oriented and object relational database.
- (c) Write short notes on :
  - (i) Intrusion detection techniques in database
  - (ii) Authorization, Authentication and access control in database



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**B022413(022)**

**B.Tech. (Fourth Semester) Examination**

**April-May 2022**

**(AICTE Scheme)**

**(CSE Branch)**

**DATABASE MANAGEMENT SYSTEM**

***Time Allowed : Three hours***

***Maximum Marks : 100***

***Minimum Pass Marks : 35***

***Note : Attempt all questions. Part (a) is compulsory from each unit. Attempt any two parts from (b), (c) and (d) in each unit. All units carries equal marks.***

**Unit-I**

1. (a) Define data independence.

4

[ 2 ]

- (b) Explain Procedural and Non procedural DML in detail. 8
- (c) Enumerate the different types of data models existing for structured data storage. 8
- (d) Explain terminology used in entity relationship model with suitable example. 8

### Unit-II

2. (a) Define Relation. 4
- (b) Define Relational algebra. What are the operation in relational algebra? 8
- (c) Describe tuple – relational calculus and domain relational calculus with various example. 8
- (d) Explain BCNF (Boyce Codd Normal Form) with example. 8

### Unit-III

3. (a) Define Query equivalence. 4
- (b) Explain basic steps of query processing with neat diagram. 8

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- (c) Explain query optimization. Explain the structure of query optimizer. 8
- (d) Write short notes on : 8
- (i) Indices
- (ii) Hashing

### Unit-IV

4. (a) What is Time Stamp? 4
- (b) What is Transaction? List the properties of transaction. 8
- (c) What do you understand by Serializability? Explain the concept of conflict serializability and view serializability with examples. 8
- (d) Explain concurrency control mechanism. 8

### Unit-V

5. (a) Define Authentication. 4
- (b) Describe MAC and RBAC models in detail. 8

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[ 4 ]

(c) Explain data warehousing with suitable diagram. 8

(d) Write short notes on : 8

(i) Distributed databases

(ii) Web databases

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**B.Tech. (Fourth Semester) Examination**

**April-May 2025**

**(CSE Branch)**

**(New Scheme)**

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***Time Allowed : Three hours***

***Maximum Marks : 100***

***Minimum Pass Marks : 35***

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**Unit-I**

- 1. (a) What is Data Independence? Why is it essential? 4**

- (b) Explain the architecture of DBMS with the help of a diagram. 8
- (c) Discuss in detail about the concepts of E-R model with suitable examples. 8
- (d) Write short notes on : 8
- (i) DDL
  - (ii) DML

## Unit-II

2. (a) List relational algebra operators with examples. 4
- (b) Implement following relation using SQL query student (stud\_no, stud\_name, sub 1, sub 2 totalmark, percentage) Create the table, add 5 records and display the data. 8
- (c) Explain the difference between open source database and commercial database. 8
- (d) What is normalization? Explain the process of relation refinement with the help of normalization by taking suitable example. (You may go up to 3 NF) 8



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### Unit-III

3. (a) What is Hashing? Explain it. 4
- (b) Does the data dictionary have any role to play in query processing? Describe with the help of an SQL query requiring JOIN operation, selection and projection. 8
- (c) How does a query tree represent a relational algebra expression? Discuss any three rules for query optimization, giving example as to when should each rule be applied. 8
- (d) What is meant by heuristic optimization? Discuss the main heuristics that are applied during query optimization. 8

### Unit-IV

4. (a) What are the ACID properties? 4
- (b) What do you understand by serializability of schedules? Explain it. 8
- (c) Explain in detail about timestamp based concurrency control techniques. 8

- (d) What is database recovery? Explain shadow paging in detail. 8

### Unit-V

5. (a) What are the various ways in which a database can be secured? 4
- (b) What are the four main access control model? 8
- (c) What are the difference between RDBMS and OODBMS? 8
- (d) What is Data Mining? Explain Data Mining Techniques. Compare between Data Mining and Data Warehousing. 8